

MUHAMMED FARIS MUKTHAR M V

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SUMMARY

Data Scientist with a Master's in Computer Science (Data Analytics) and a strong foundation in mathematics (BSc), specialising in building data pipelines, predictive models, and analytics dashboards that drive actionable decisions. Experienced in collecting and structuring real-world datasets from IoT systems, computer vision outputs, and document AI pipelines. Skilled in Python, SQL, and ML frameworks with hands-on experience delivering stakeholder insights through visual dashboards (Tableau, Power BI) and cloud-hosted applications (AWS, Supabase).

TECHNICAL SKILLS

Data Science: Predictive Modelling, EDA, Time Series Analysis, Statistical Modelling, Data Visualisation

ML / AI: YOLOv8, LayoutLM, RoBERTa, CNN/RNN/LSTM, LLMs, LangChain, RAG, Pgvector

BI & Analytics: Tableau, Power BI, Microsoft Excel, SQL, Pandas, NumPy

Libraries: Scikit-learn, PyTorch, TensorFlow, OpenCV

Cloud & Data: AWS (EC2, S3), Supabase (PostgreSQL), Microsoft Azure, Docker

Web & Mobile: Next.js, Node.js, REST APIs

Languages: Python, SQL, JavaScript, HTML/CSS, Embedded C/C++

EXPERIENCE

R&D Data Scientist – AI & Innovations

Oct 2025 – Present

Digital University Kerala

Technopark, Trivandrum

- Built a full-stack physical education management system combining IoT hardware, computer vision, and AI analytics for Kerala state schools.
- Designed and managed a **data collection and analytics pipeline** for a children's health monitoring platform — aggregating IoT sensor readings, computer vision activity metrics, and teacher feedback into a structured PostgreSQL (Supabase) dataset.
- Performed **student performance analysis** on the collected dataset, identifying trends and improvement patterns; delivered visual insights to super admins and role-specific stakeholders via a web dashboard (Next.js, AWS).
- Deployed **YOLOv8 computer vision** on Raspberry Pi for automated activity detection and performance measurement, feeding structured results into the analytics pipeline.
- Built a **hierarchical multi-user web application** (Next.js, Supabase/PostgreSQL, AWS EC2/S3) with role-based dashboards for super admins, teachers, and students enabling data-driven drill assignment and progress tracking.
- Developing a **RAG-based student health chatbot** using LangChain and LLMs grounded in the collected health dataset, providing personalised, data-driven recommendations.
- Designed database schemas and REST APIs (Node.js) to support structured data ingestion from IoT hardware, mobile apps, and computer vision inference pipelines.
- Mentored students on data science project formulation, ML model selection, and applied analytics for hackathons.

R&D Intern – AI & Data Systems

Apr 2025 – Oct 2025

Cisco thinQbator-DUK

Technopark, Trivandrum

- Built real-time sensor data visualisation dashboards in Next.js displaying live IoT readings and ML predictions.
- Developed structured IoT data pipelines using Raspberry Pi, Arduino, and ESP32 feeding clean data to backend services.
- Integrated ML models with edge hardware for real-time classification and detection tasks.

Consultant – Machine Learning Engineering

Sep 2024 – Dec 2024

Cleareye.ai

Technopark Phase I, Trivandrum

- Fine-tuned **RoBERTa** for entity detection in Trade Finance documents and built **LayoutLM + OpenCV** pipelines for layout analysis, significantly improving extraction accuracy and reducing manual effort.
- Implemented **Siamese networks** for signature verification and **YOLOv8** for document seal detection in production workflows.
- Created **synthetic datasets** to resolve data scarcity, improving model generalisation and training quality.

Machine Learning Intern

Cleareye.ai

May 2024 – Aug 2024

Technopark Phase I, Trivandrum

- Fine-tuned LayoutLM for domain-specific document extraction and built content/metadata classification pipelines.
- Generated synthetic training datasets to expand sample size and improve model robustness.

Machine Learning Research Intern

Digital University Kerala

Feb 2024 – May 2024

Technopark, Trivandrum

- Performed document sanctity checks using structural OCR and LayoutLM layout analysis models.
- Automated seal/signature detection with YOLO, reducing manual verification effort substantially.

PROJECTS

Customer Retail Trend & Time Series Forecasting | *ML, Time Series, Scikit-learn*

- Analysed purchasing trends and built a predictive purchase model delivering actionable business insights for customer satisfaction and revenue growth decisions.

Document Extraction Chatbot | *LLMs, RAG, LangChain, TensorFlow*

- Government order chatbot with LangChain RAG pipeline for context-aware query resolution over large document collections.

Plant Disease Detection | *YOLOv8, OpenCV, Python*

- Real-time leaf image classification using YOLOv8 to detect plant diseases and surface actionable agricultural insights.

EDUCATION

Master of Computer Science in Data Analytics

Digital University Kerala

Aug 2022 – Jul 2024

Trivandrum, Kerala

Bachelor of Science in Mathematics

Farook College, Calicut University

Jun 2019 – Mar 2022

Calicut, Kerala

ACTIVITIES

UGC-NET — Qualified for admission to Ph.D. ONLY

2026

NCC 'C' Certificate Holder

2022

National Cadet Corps (NCC), India — Company Quartermaster Sergeant

2020 – 2022